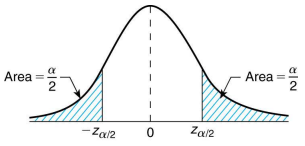
Agenda

- Earlier
  - Confidence intervals for two populations or treatments
- Today
  - Hypothesis testing (two populations)

In India, Gallup puts approval at 36%, exceeding disapproval by at least 10 percentage points. Pew found that 40% of Indians viewed UN favourably, and only 18% unfavourably

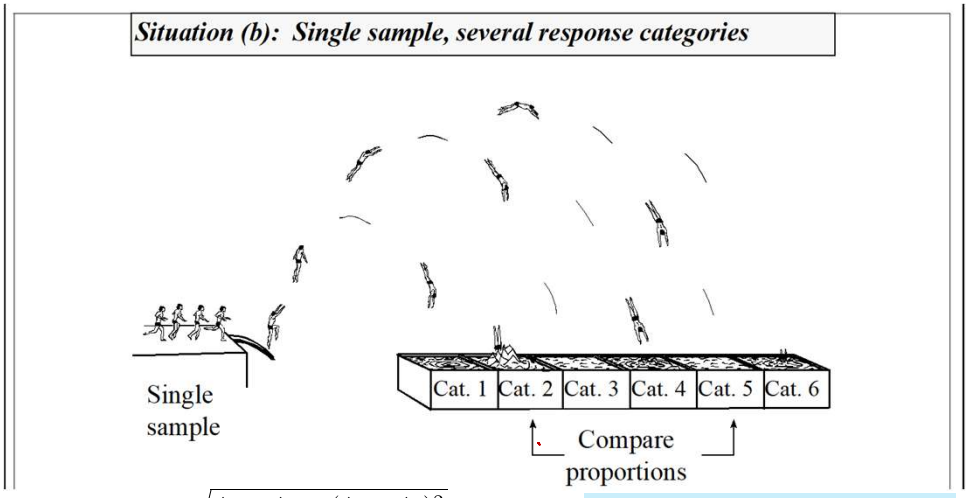


1

Review

2

Proportions



$$\text{se}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1 + \hat{p}_2 - (\hat{p}_1 - \hat{p}_2)^2}{n}}$$

Want to estimate true difference in the proportions $\widehat{P_1} - \widehat{P_2}$

3

Proportions for Case (b)

- Sample n independent people with given proportions
- Observe two mutually exclusive indicators for each person
  - $X_i = \mathbf{1}\{\text{person i chooses characteristic 1}\}$, and $Y_i = \mathbf{1}\{\text{person i chooses characteristic 2}\}$
  - $\mathbb{E}[X_i] = p_1$ (also probability that person i will choose characteristic 1)
  - Define: Person level difference as $D_i = X_i - Y_i$
  - $D_i = 1$ with probability p_1, -1 with probability p_2 and 0 with probability $1 - p_1 - p_2$
  - $E[D_i] = p_1 - p_2, E[D_i^2] = p_1 + p_2, \text{Var}(D_i) = p_1 + p_2 - (p_1 - p_2)^2$
- For the entire population,
  - Define $D = \sum_i^n D_i$, then difference P between the sample proportions is D/n
  - Applying the CLT approximation we have $P \overset{\text{approx}}{\sim} \mathcal{N}\left(p_1 - p_2, \frac{p_1 + p_2 - (p_1 - p_2)^2}{n}\right)$

4

Significance Testing via Hypothesis

Warning: concepts coming up, not proofs

5

Review: Hypothesis Testing

- We cannot “rule-in” a hypothesized value for a parameter
- We can only determine whether there is evidence to rule out a hypothesized value

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Review: Hypothesis Testing

- In settling a research hypothesis, we turn skeptical and assume the research hypothesis is false
  - The null hypothesis is H_0 (or H_o)
  - The alternate is the research hypothesis H_1 or H_a
- Is there data to reject the null hypothesis?
  - Solution:
    - Compute the test statistic
      - Is this statistic "too far" from the null hypothesis?
    - If true, then reject the null hypothesis
    - If false, cannot reject the null hypothesis

$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

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Review: The t-test

- Using $\hat{\theta}$ to test $H_0: \theta = \theta_0$ versus H_a
  1. Calculate the test statistic t_0
  2. Calculate the p-value
  3. Interpret the p-value

These numbers are "rule of thumb"

P	%	Translation
> 0.12	-12%	No evidence against H0
0.1	-10%	Weak evidence against H0
0.05	-5%	Some evidence against H0
0.01	-1%	Strong evidence against H0
0.001	-0.10%	Very Strong evidence against H0

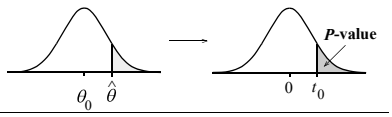
$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

Alternative hypothesis	Evidence against H_0: $\theta > \theta_0$ provided by	P-value
$H_1: \theta > \theta_0$	too much bigger than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too large)	$P = pr(T \geq t_0)$
$H_1: \theta < \theta_0$	too much smaller than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too negative)	$P = pr(T \leq t_0)$
$H_1: \theta \neq \theta_0$	too far from θ_0 (i.e., $\lvert\hat{\Theta} - \theta_0\rvert$ too large)	$P = 2\, pr(T \geq \lvert t_0 \rvert)$
		where T ~ Student(df)

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P-value from t-tests

- The P-value is the probability that, if the hypothesis was true, sampling variation would produce an estimate that is further away from the hypothesized value than our data-estimate
- The P-value measures the strength of the evidence **against** H_0
- The **smaller** the P-value, the **stronger** the evidence against H_0
  - The second and third points are true for significance tests generally, and not just for t-tests

Alternative Hypothesis	Evidence against H_0: $\theta = \theta_0$ provided by	$\hat{\theta}$-scale → *t*-scale (# of std errors)
H_1: $\theta > \theta_0$	$\hat{\theta}$ too much bigger than θ_0	

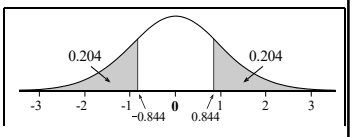
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Example #2 Piazza post 242

$$t_o = \frac{\hat{\theta} - \theta_0}{\text{se}(\hat{\theta})}$$

- In 1798, Cavendish measured the density of earth
  - After 6 experiments, he had to change his equipment resulting in the data
  - n = 23, sample mean = 5.483, sample sd = 0.1904 (Accepted value 5.517)
  - Research hypothesis: His last 23 experiments resulted in biased measurement
    - Null hypothesis: H_0: His experiments were not biased, the data followed the normal distribution
      - We do not know the mean and variance of the distribution corresponding to H_0
      - But we have sample variance, and sample standard deviation
    - Compute the t-statistic: -0.844, Compute the p-value: 0.204
    - Interpretation: Cannot refute H_o

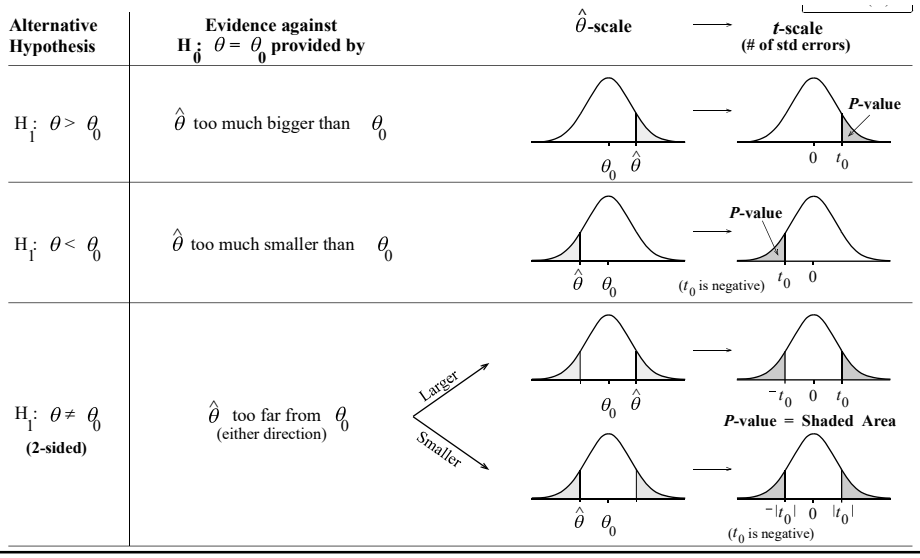
Piazza post 242: Why in the cavendish example did we take t distribution t



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Clarification: p-value is not 0.204 but 0.408

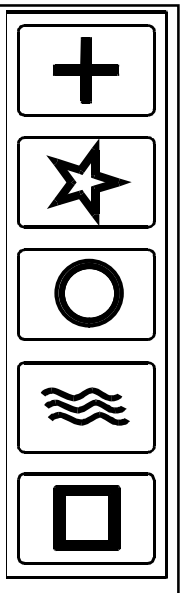
The t-test



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Example #1: Piazza post 154

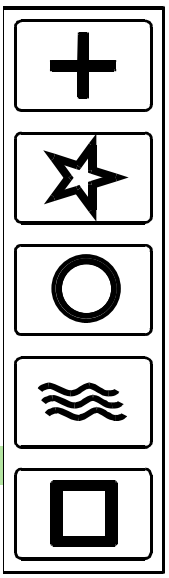
- In 1938, scientists showed 60,000 pictures
  - Students predict what is behind the card
  - If students guessed randomly there would be 12000 correct answers (proportion is 0.2)
  - … but actual result was 12489 (proportion is 0.208)
  - Is 20.8% being larger than 20% a "chance event" or
  - Hypothesis: Students can mind read correct answer in an MCQ



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Example #1: Piazza post 154

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$



- In 1938, scientists collected 60,000 pieces of data
  - If students guessed randomly
    - there would be 12000 correct answers, but actual result was 12489
  - Research hypothesis: Students mind read correct answer in an MCQ
  - H_0: the experiment is Bernoulli (0.2)
    - For n = 60000, we apply the normal approximation to the proportion
    - We know the mean and variance (piazza post 154)
      - t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001633, t=4.99
      - OR t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001657, t=4.92

Piazza post 154: Take this

    - Either way: p-value: < 10^{-6}
    - Interpret: Can refute the null hypothesis
    - Inference: Not pure guessing (maybe mind read?)

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p-value for proportions

Piazza post 154: Take this

- Which form to use for the p-value?
  - t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001633, t=4.99
  - OR t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001657, t=4.92

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

- There are (at least) 3 ways all asymptotically equivalent
  1. Lagrange multiplier test or Score test (mostly taught in intro stats, more principled in this case, piazza post 154)
  2. Likelihood Ratio Test
  3. Wald test (Oct05 class): *corresponds to the confidence interval test*
  - (One can of course use the exact test: just use the binomial)

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Type I Errors

- If the null hypothesis is true, and we decide to reject
  - at the 95% confidence level, we will make an error 5% of the time
  - At the 99% level, we would be less wrong
  - Never reject, we would never be wrong

Decision Making	
	Actual situation
Decision made	H_0 is true
Do not reject H0	OK
Reject H_0 as false	Type I error

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Tests and Intervals

- Two different methods for coping with the uncertainty
  - True value of a parameter caused by the sampling variation in estimates
  - Confidence interval: A fixed level of confidence is chosen. We determine a range of possible values for the parameter that are consistent with the data
  - Significance test: Only one possible value for the parameter, called the hypothesized value, is tested. We determine the strength of the evidence provided by the data against the proposition that the hypothesized value is the true value
- The significance level is the probability of making a Type-I error
  - A two-sided test of $H_0: \theta = \theta_0$ is significant at the 5% level if and only if θ_0 lies outside a 95% confidence interval for θ

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Tests and Intervals

Must know

- Statistical significance relates to the existence of an effect.
- The practical significance of an effect depends on its size.
- A small P-value provides evidence that the effect exists but says nothing at all about the size of the effect.
  - Example: Risk of death for smokers prescribed beta-blockers
    - P-value = 0.048 (Cannot reject null hypothesis "beta-blockers have no effect on death")
    - Confidence interval (factor) [1.004, 2.5]
- To estimate the size of an effect, compute a confidence interval
  - Never quote a P-value about the existence of an effect without also providing a confidence interval estimating its size.

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Clarification:
The sentence in quotes is H_0.
p-value at $\alpha = 0.05$ fails to reject H_0